

## 4.4 Independence of quadratic forms and linear forms

An  $F$  or  $t$  statistic is a ratio, and its distribution is known only if numerator and denominator are independent. For normal vectors, uncorrelated linear functions are independent (Theorem 3.4.1). Quadratic forms are not linear, but the following observation brings them within reach of that result.

*Idea.* A quadratic form is a function of a linear form: for any generalized inverse  $\mathbf{A}^-$  of  $\mathbf{A}$ ,

$$\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = \mathbf{Y}^\top \mathbf{A} \mathbf{A}^- \mathbf{A} \mathbf{Y} = (\mathbf{A} \mathbf{Y})^\top \mathbf{A}^- (\mathbf{A} \mathbf{Y}).$$

So  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  is independent of anything that is independent of the vector  $\mathbf{A} \mathbf{Y}$ .

### 4.4.1 The independence theorems

#### Theorem 4.4.1 — Linear and quadratic forms

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ , let  $\mathbf{B}$  be  $m \times n$  and  $\mathbf{A}$  symmetric  $n \times n$ . If  $\mathbf{B} \boldsymbol{\Sigma} \mathbf{A} = \mathbf{0}$ , then  $\mathbf{B} \mathbf{Y}$  and  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  are independent. Conversely, if  $\mathbf{B} \mathbf{Y}$  and  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  are independent for every  $\boldsymbol{\mu} \in \mathbb{R}^n$ , with  $\boldsymbol{\Sigma}$  fixed, then  $\mathbf{B} \boldsymbol{\Sigma} \mathbf{A} = \mathbf{0}$ .

*Proof.* The vector  $(\mathbf{B} \mathbf{Y}, \mathbf{A} \mathbf{Y})$  is a linear function of  $\mathbf{Y}$ , hence jointly normal (Theorem 3.3.1), and its cross-covariance is  $\text{Cov}(\mathbf{B} \mathbf{Y}, \mathbf{A} \mathbf{Y}) = \mathbf{B} \boldsymbol{\Sigma} \mathbf{A}^\top = \mathbf{B} \boldsymbol{\Sigma} \mathbf{A}$  (Theorem 2.2.1). If this is zero,  $\mathbf{B} \mathbf{Y}$  and  $\mathbf{A} \mathbf{Y}$  are independent by Theorem 3.4.1, and so are  $\mathbf{B} \mathbf{Y}$  and the function  $(\mathbf{A} \mathbf{Y})^\top \mathbf{A}^- (\mathbf{A} \mathbf{Y})$  of  $\mathbf{A} \mathbf{Y}$ , which is  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  by the key idea above. Conversely, independent variables with finite variances are uncorrelated, so Theorem 4.2.3(c) gives  $2\mathbf{B} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu} = \mathbf{0}$  for every  $\boldsymbol{\mu}$ , that is,  $\mathbf{B} \boldsymbol{\Sigma} \mathbf{A} = \mathbf{0}$ . ■

#### Theorem 4.4.2 — Two quadratic forms

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  and let  $\mathbf{A}, \mathbf{B}$  be symmetric. If  $\mathbf{A} \boldsymbol{\Sigma} \mathbf{B} = \mathbf{0}$ , equivalently  $\mathbf{B} \boldsymbol{\Sigma} \mathbf{A} = \mathbf{0}$ , then  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  and  $\mathbf{Y}^\top \mathbf{B} \mathbf{Y}$  are independent.

*Proof.* The two conditions are transposes of each other. As before,  $\mathbf{A} \mathbf{Y}$  and  $\mathbf{B} \mathbf{Y}$  are jointly normal with  $\text{Cov}(\mathbf{A} \mathbf{Y}, \mathbf{B} \mathbf{Y}) = \mathbf{A} \boldsymbol{\Sigma} \mathbf{B} = \mathbf{0}$ , hence independent, and each form is a function of one of them. ■

The condition is sufficient for every  $\boldsymbol{\mu}$  and every  $\boldsymbol{\Sigma}$ . When  $\boldsymbol{\Sigma}$  is singular one might hope that the weaker requirement  $\boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} \mathbf{B} \boldsymbol{\Sigma} = \mathbf{0}$  suffices, since only directions in  $\mathcal{C}(\boldsymbol{\Sigma})$  carry noise. For nonnegative definite  $\mathbf{A}$  and  $\mathbf{B}$  the two requirements turn out to be equivalent (Exercise (B2)). In the other direction, the condition is necessary under mild assumptions, a result usually credited

to Craig (1943). Its general proof is surprisingly delicate, and the history of incorrect published proofs is recounted by Driscoll and Gundberg (1986). The case that matters most for sums of squares has a short proof.

### Proposition 4.4.3 — Necessity for nonnegative definite forms

Let  $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$  and let  $\mathbf{A}, \mathbf{B}$  be nonnegative definite. If  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  and  $\mathbf{Y}^\top \mathbf{B} \mathbf{Y}$  are independent, then  $\mathbf{A} \mathbf{B} = \mathbf{0}$ .

*Proof.* Independence makes the covariance zero, and by Theorem 4.2.3(d) the covariance is  $2 \operatorname{tr}(\mathbf{A} \mathbf{B})$ . With the square roots of Theorem 1.7.6,

$$\operatorname{tr}(\mathbf{A} \mathbf{B}) = \operatorname{tr}(\mathbf{A}^{1/2} \mathbf{A}^{1/2} \mathbf{B}^{1/2} \mathbf{B}^{1/2}) = \operatorname{tr}\{(\mathbf{A}^{1/2} \mathbf{B}^{1/2})^\top (\mathbf{A}^{1/2} \mathbf{B}^{1/2})\},$$

the sum of squares of the entries of  $\mathbf{A}^{1/2} \mathbf{B}^{1/2}$ . So  $\mathbf{A}^{1/2} \mathbf{B}^{1/2} = \mathbf{0}$ , and  $\mathbf{A} \mathbf{B} = \mathbf{A}^{1/2} (\mathbf{A}^{1/2} \mathbf{B}^{1/2}) \mathbf{B}^{1/2} = \mathbf{0}$ . ■

For nonnegative definite forms in a standard normal vector, then, zero correlation and independence coincide, just as for jointly normal linear forms. This fails for general symmetric matrices (Exercise (B1)).

## 4.4.2 Orthogonal projections of a spherical normal vector

The situation in regression is the tidiest one: a spherical normal vector and several projections onto mutually orthogonal subspaces.

### Theorem 4.4.4 — Independent projections

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$ , and let  $\mathbf{P}_1, \dots, \mathbf{P}_k$  be symmetric idempotent matrices with  $\mathbf{P}_i \mathbf{P}_j = \mathbf{0}$  for  $i \neq j$ , and ranks  $r_1, \dots, r_k$ . Then  $\mathbf{P}_1 \mathbf{Y}, \dots, \mathbf{P}_k \mathbf{Y}$  are mutually independent normal vectors,  $\mathbf{P}_i \mathbf{Y} \sim N_n(\mathbf{P}_i \boldsymbol{\mu}, \sigma^2 \mathbf{P}_i)$ , and the sums of squares

$$\frac{\|\mathbf{P}_i \mathbf{Y}\|^2}{\sigma^2} \sim \chi^2\left(r_i, \frac{\|\mathbf{P}_i \boldsymbol{\mu}\|^2}{\sigma^2}\right), \quad i = 1, \dots, k,$$

are mutually independent.

*Proof.* Stack the vectors  $\mathbf{P}_i \mathbf{Y}$  into one  $nk$ -vector. It is a linear function of  $\mathbf{Y}$ , so it is normal, and the covariance between blocks  $i$  and  $j$  is  $\mathbf{P}_i (\sigma^2 \mathbf{I}) \mathbf{P}_j^\top = \sigma^2 \mathbf{P}_i \mathbf{P}_j$ . This is  $\mathbf{0}$  for  $i \neq j$  and  $\sigma^2 \mathbf{P}_i$  for  $i = j$ . By the block form of Theorem 3.4.1 the blocks are mutually independent, and functions of them, the squared lengths, are then mutually independent. Their laws are Theorem 4.3.2(a). ■

In geometric language, which [Chapter 6](#) develops: the coordinates of a spherical normal vector in mutually orthogonal subspaces are independent. It is the reason the rows of an analysis of variance table can be treated separately.

### Corollary 4.4.5 — Sample mean and sample variance

Let  $Y_1, \dots, Y_n$  ( $n \geq 2$ ) be independent  $N(\mu, \sigma^2)$ , with sample mean  $\bar{Y}$  and sample variance  $S^2 = (n-1)^{-1} \sum_i (Y_i - \bar{Y})^2$ . Then

- (a)  $\bar{Y}$  and  $S^2$  are independent, and  $\bar{Y} \sim N(\mu, \sigma^2/n)$ ;
- (b)  $(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$ ;
- (c) for any  $\mu_0$ ,  $T = \sqrt{n}(\bar{Y} - \mu_0)/S \sim t(n-1, \delta)$  with  $\delta = \sqrt{n}(\mu - \mu_0)/\sigma$ . In particular  $T \sim t(n-1)$  when  $\mu = \mu_0$ .

*Proof.* Here  $\mathbf{Y} \sim N_n(\mu \mathbf{1}, \sigma^2 \mathbf{I})$ . Let  $\mathbf{P}_1 = n^{-1} \mathbf{1} \mathbf{1}^\top$  and  $\mathbf{C} = \mathbf{I} - \mathbf{P}_1$ . Both are symmetric and idempotent,  $\mathbf{P}_1 \mathbf{C} = \mathbf{0}$ , and  $\text{rank}(\mathbf{C}) = \text{tr}(\mathbf{C}) = n-1$ . Since  $\mathbf{P}_1 \mathbf{Y} = \bar{Y} \mathbf{1}$  and  $\|\mathbf{C} \mathbf{Y}\|^2 = (n-1)S^2$ , [Theorem 4.4.4](#) gives (a), which Example (Sample mean and deviations) obtained directly, and, because  $\mathbf{C} \mu \mathbf{1} = \mathbf{0}$ , also (b). For (c),  $X = \sqrt{n}(\bar{Y} - \mu_0)/\sigma \sim N(\delta, 1)$  and  $V = (n-1)S^2/\sigma^2 \sim \chi^2(n-1)$  are independent, and  $T = X/\sqrt{V/(n-1)}$ . Apply Definition 4.1.3. ■

The independence in (a) characterizes the normal distribution: for  $n \geq 2$  independent identically distributed observations with finite variance,  $\bar{Y}$  and  $S^2$  are independent only under normality (Lukacs 1942). We will not need this, but it shows that normality is used essentially here. It is not a technicality that more careful arguments could remove.

## 4.4.3 Preview: the $F$ statistic for nested models

The main use of these results comes in [Chapter 11](#). Its core fits in a few lines.

### Theorem 4.4.6 — $F$ statistic for nested subspaces

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$ , and let  $\mathbf{P}_0$  and  $\mathbf{P}$  be symmetric idempotent matrices of ranks  $r_0 < r < n$  with  $\mathbf{P} \mathbf{P}_0 = \mathbf{P}_0$ . Suppose that  $\mathbf{P} \boldsymbol{\mu} = \boldsymbol{\mu}$ . Then

$$F = \frac{\mathbf{Y}^\top (\mathbf{P} - \mathbf{P}_0) \mathbf{Y} / (r - r_0)}{\mathbf{Y}^\top (\mathbf{I} - \mathbf{P}) \mathbf{Y} / (n - r)} \sim F\left(r - r_0, n - r, \frac{\|(\mathbf{P} - \mathbf{P}_0) \boldsymbol{\mu}\|^2}{\sigma^2}\right),$$

and the noncentrality is zero iff  $\mathbf{P}_0 \boldsymbol{\mu} = \boldsymbol{\mu}$ .

*Proof.* Transposing  $\mathbf{P} \mathbf{P}_0 = \mathbf{P}_0$  gives  $\mathbf{P}_0 \mathbf{P} = \mathbf{P}_0$ . Hence  $(\mathbf{P} - \mathbf{P}_0)^2 = \mathbf{P} - 2\mathbf{P}_0 + \mathbf{P}_0 = \mathbf{P} - \mathbf{P}_0$ , so  $\mathbf{P} - \mathbf{P}_0$  is symmetric idempotent with rank  $\text{tr}(\mathbf{P}) - \text{tr}(\mathbf{P}_0) = r - r_0$ . Also  $(\mathbf{P} - \mathbf{P}_0)(\mathbf{I} - \mathbf{P}) = \mathbf{P} - \mathbf{P} - \mathbf{P}_0 + \mathbf{P}_0 = \mathbf{0}$ , and  $\mathbf{I} - \mathbf{P}$  is symmetric idempotent of rank  $n - r$ . By [Theorem 4.4.4](#) the numerator and denominator sums of squares are independent, with  $\sigma^{-2}$  times them distributed as  $\chi^2(r - r_0, \|(\mathbf{P} - \mathbf{P}_0) \boldsymbol{\mu}\|^2 / \sigma^2)$  and  $\chi^2(n - r, \|(\mathbf{I} - \mathbf{P}) \boldsymbol{\mu}\|^2 / \sigma^2)$ . The second noncentrality is 0 since  $\mathbf{P} \boldsymbol{\mu} = \boldsymbol{\mu}$ . The unknown  $\sigma^2$  cancels in the ratio, and

Definition 4.1.2 applies. Finally,  $(\mathbf{P} - \mathbf{P}_0)\boldsymbol{\mu} = \boldsymbol{\mu} - \mathbf{P}_0\boldsymbol{\mu}$ .

■

In a linear model,  $\mathbf{P} = \mathbf{M}$  projects onto the column space of the full model matrix  $\mathbf{X}$  and  $\mathbf{P}_0 = \mathbf{M}_0$  onto that of a submodel  $\mathbf{X}_0$ . The numerator is the drop in residual sum of squares from the submodel to the full model (Chapter 6). The theorem says three things. The test that rejects for large  $F$  has an exactly known null distribution. Its power depends on the unknown parameters only through  $\gamma = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2$ . And, by Theorem 4.1.4, that power increases with  $\gamma$ .

### Example. (Detecting curvature)

Let  $x_1, \dots, x_n$ ,  $n = 15$ , be equally spaced on  $[0, 1]$ , and suppose  $Y_i = 1 + 0.5x_i + \beta_2 x_i^2 + \varepsilon_i$  with independent  $N(0, 1)$  errors. To test for curvature, compare the straight line ( $\mathbf{X}_0 = [\mathbf{1}, \mathbf{x}]$ ) with the quadratic ( $\mathbf{X} = [\mathbf{1}, \mathbf{x}, \mathbf{x}^2]$ ). The  $F$  statistic has 1 and 12 degrees of freedom, and the 5% critical value is 4.747. For  $\beta_2 = 8.0$ , the listing finds  $\gamma = 6.873$  and power 0.673 from the noncentral  $F$ , and the rejection rate in 100,000 simulated data sets is 0.674.

The noncentrality has a clear meaning. It is  $\beta_2^2$  times the squared length of the part of  $\mathbf{x}^2$  orthogonal to  $\mathbf{1}$  and  $\mathbf{x}$ , divided by  $\sigma^2$ . It grows roughly in proportion to  $n$  for a fixed design pattern. Keeping the same design on  $[0, 1]$ , the smallest  $n$  with power at least 0.8 is 21, where  $\gamma = 8.97$ . Figure 4.4.1 places the example on the power curve and shows the loss of power when the same noncentrality is spread over more numerator degrees of freedom  $r$ . At  $\gamma = 10$  the power is 0.827, 0.606 and 0.421 for  $r = 1, 3, 6$ .

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(1104)
alpha, s = 0.05, 12
def power(gamma, q, s, alpha=0.05):
    crit = stats.f.ppf(1 - alpha, q, s)
    return stats.ncf.sf(crit, q, s, np.maximum(gamma, 1e-12))
gammas = np.linspace(0, 30, 301)
gamma_ref, rs = 10.0, (1, 3, 6) # r = numerator degrees of
                                # freedom
curves = {q: power(gammas, q, s) for q in rs}

n, sigma, beta0, beta1, beta2, reps = 15, 1.0, 1.0, 0.5, 8.0, 100_000
x = np.linspace(0, 1, n)
X0 = np.column_stack([np.ones(n), x]) # straight line
X = np.column_stack([X0, x**2]) # adds curvature

def proj(Z):
    Q, _ = np.linalg.qr(Z)
    return Q @ Q.T

M, M0 = proj(X), proj(X0)
mean = X @ np.array([beta0, beta1, beta2])
gamma = mean @ (M - M0) @ mean / sigma**2 # noncentrality
q, df_resid = 1, n - 3
crit = stats.f.ppf(0.95, q, df_resid)
exact_power = stats.ncf.sf(crit, q, df_resid, gamma)
```

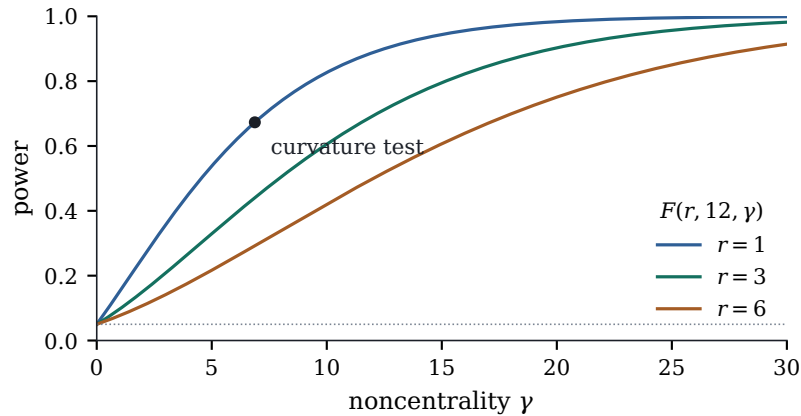


Figure 4.4.1: Power  $\Pr\{F(r, 12, \gamma) > F_{0.05}(r, 12)\}$  of the level-0.05  $F$  test as a function of the noncentrality. Every curve starts at 0.05 and increases to 1 (Theorem 4.1.4). For a fixed  $\gamma$ , power falls as the number  $r$  of numerator degrees of freedom grows. The dot is the curvature test of [Example \(Detecting curvature\)](#).

```
Y = mean + sigma * rng.standard_normal((reps, n))
num = np.einsum("ij,jk,ik->i", Y, M - M0, Y) / q
den = np.einsum("ij,jk,ik->i", Y, np.eye(n) - M, Y) / df_resid
print(gamma, exact_power, np.mean(num / den > crit))
```

[Run this code online](#)

## 4.4.4 Exercises

### 4.4.4.1 A. Check your understanding

#### Exercise A1.

In Example (Two estimators of  $\sigma^2$  under a drifting mean) let  $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \sigma^2\mathbf{I})$  with  $n \geq 3$ . Use Theorem 4.2.3(d) to show that  $\text{Cov}(S^2, Q) = \text{Var}(S^2)$ , and deduce that  $S^2$  and  $Q$  are dependent with correlation  $\{2(n-1)/(3n-4)\}^{1/2}$ . Which linear forms  $\mathbf{a}^\top \mathbf{Y}$  are independent of  $Q$  for every mean vector  $\mu \in \mathbb{R}^n$ ?

*Solution.* With  $\Sigma = \sigma^2\mathbf{I}$ , part (d) of Theorem 4.2.3 gives  $\text{Cov}(S^2, Q) = 2\sigma^4 \text{tr}(\mathbf{C}\mathbf{D}^\top \mathbf{D}) / \{2(n-1)^2\}$ . Since  $\mathbf{D}\mathbf{1} = \mathbf{0}$ ,  $\mathbf{C}\mathbf{D}^\top \mathbf{D} = \mathbf{D}^\top \mathbf{D}$ , whose trace is  $2(n-1)$ , so  $\text{Cov}(S^2, Q) = 2\sigma^4 / (n-1) = \text{Var}(S^2)$ . The mean terms vanish because  $\mathbf{C}\mu = \mathbf{D}\mu = \mathbf{0}$ . Using  $\text{Var}(Q)$  from Example, the correlation is  $\{\text{Var}(S^2) / \text{Var}(Q)\}^{1/2} = \{2(n-1)/(3n-4)\}^{1/2}$ , which is nonzero, so the forms are dependent. By [Theorem 4.4.1](#),  $\mathbf{a}^\top \mathbf{Y}$  is independent of  $Q$  for every  $\mu$  iff  $\mathbf{a}^\top \mathbf{D}^\top \mathbf{D} = \mathbf{0}^\top$ . Multiplying by  $\mathbf{a}$  gives  $\|\mathbf{D}\mathbf{a}\|^2 = 0$ , so  $\mathbf{D}\mathbf{a} = \mathbf{0}$  and  $\mathbf{a}$  is a multiple of  $\mathbf{1}$ ; conversely such  $\mathbf{a}$  work. The only such linear forms are the multiples of  $\bar{Y}$ .

### 4.4.4.2 B. Practice

**Exercise B1.**

Let  $\mathbf{Y} \sim N_2(\mathbf{0}, \mathbf{I})$ . Show that  $Y_1^2 - Y_2^2$  and  $2Y_1Y_2$  are uncorrelated but not independent. Why does Proposition 4.4.3 not apply?

*Solution.* With  $\mathbf{A} = \text{diag}(1, -1)$  and  $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ ,  $\text{tr}(\mathbf{AB}) = 0$ , so the covariance  $2\text{tr}(\mathbf{AB})$  is zero. In polar coordinates  $Y_1 = R \cos \Theta$ ,  $Y_2 = R \sin \Theta$ , with  $R^2 \sim \chi^2(2)$  independent of  $\Theta$  uniform on  $[0, 2\pi)$ , the two forms are  $R^2 \cos 2\Theta$  and  $R^2 \sin 2\Theta$ . Now  $E(R^4) = 8$  and  $E(R^8) = 384$ , so  $E[(R^2 \cos 2\Theta)^2 (R^2 \sin 2\Theta)^2] = 384/8 = 48$ , but the product of the expected squares is  $(8/2)^2 = 16$ . The forms are dependent. Proposition 4.4.3 needs nonnegative definite matrices, and  $\mathbf{A}$  and  $\mathbf{B}$  are indefinite.

**Exercise B2.**

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  and let  $\mathbf{A}, \mathbf{B}$  be nonnegative definite with  $\boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} \mathbf{B} \boldsymbol{\Sigma} = \mathbf{0}$ . Show that  $\mathbf{A} \boldsymbol{\Sigma} \mathbf{B} = \mathbf{0}$ , so that  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  and  $\mathbf{Y}^\top \mathbf{B} \mathbf{Y}$  are independent. Show by an example that the implication can fail when  $\mathbf{A}$  is not nonnegative definite.

*Solution.* Write  $\mathbf{A} = \mathbf{R} \mathbf{R}^\top$  and  $\mathbf{B} = \mathbf{S} \mathbf{S}^\top$ , for instance with symmetric square roots, and  $\boldsymbol{\Sigma} = \mathbf{L} \mathbf{L}^\top$  with  $\mathbf{L}$  of full column rank. Cancelling  $\mathbf{L}$  on both sides of  $\mathbf{L}(\mathbf{L}^\top \mathbf{A} \mathbf{L})(\mathbf{L}^\top \mathbf{B} \mathbf{L})\mathbf{L}^\top = \mathbf{0}$  gives  $\mathbf{L}^\top \mathbf{A} \mathbf{L} \mathbf{L}^\top \mathbf{B} \mathbf{L} = \mathbf{0}$ . The trace of this matrix is

$$\text{tr}(\mathbf{L}^\top \mathbf{R} \mathbf{R}^\top \mathbf{L} \mathbf{L}^\top \mathbf{S} \mathbf{S}^\top \mathbf{L}) = \text{tr}\{(\mathbf{R}^\top \boldsymbol{\Sigma} \mathbf{S})(\mathbf{R}^\top \boldsymbol{\Sigma} \mathbf{S})^\top\} = \|\mathbf{R}^\top \boldsymbol{\Sigma} \mathbf{S}\|_F^2,$$

so  $\mathbf{R}^\top \boldsymbol{\Sigma} \mathbf{S} = \mathbf{0}$ . Hence  $\mathbf{A} \boldsymbol{\Sigma} \mathbf{B} = \mathbf{R}(\mathbf{R}^\top \boldsymbol{\Sigma} \mathbf{S})\mathbf{S}^\top = \mathbf{0}$ , and Theorem 4.4.2 applies. For nonnegative definite matrices the two conditions are therefore equivalent, since  $\mathbf{A} \boldsymbol{\Sigma} \mathbf{B} = \mathbf{0}$  trivially implies  $\boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} \mathbf{B} \boldsymbol{\Sigma} = \mathbf{0}$ . Without nonnegative definiteness the argument breaks at the trace step. Take  $\boldsymbol{\Sigma} = \text{diag}(1, 0)$ ,  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  and  $\mathbf{B} = \mathbf{I}_2$ . Then  $\boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} = \mathbf{0}$ , so  $\boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} \mathbf{B} \boldsymbol{\Sigma} = \mathbf{0}$ , but  $\mathbf{A} \boldsymbol{\Sigma} \mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \neq \mathbf{0}$ . With  $\boldsymbol{\mu} = (0, m)^\top$  and  $m \neq 0$  we have  $Y_2 = m$  with probability one, so the forms are  $2mY_1$  and  $Y_1^2 + m^2$ . The second is a nonconstant function of the first, so they are dependent.

**Exercise B3.**

Let  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$  with  $\mathbf{X}$  of full column rank  $p < n$ , let  $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ ,  $\mathbf{M} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$  and  $s^2 = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2 / (n - p)$ . With  $v_{jj}$  the  $j$ th diagonal entry of  $(\mathbf{X}^\top \mathbf{X})^{-1}$ , show that  $(\hat{\beta}_j - b) / (s\sqrt{v_{jj}}) \sim t(n - p, (\beta_j - b) / (\sigma\sqrt{v_{jj}}))$ .

*Solution.* Here  $\mathbf{M}$  is symmetric and idempotent with  $\text{tr} \mathbf{M} = \text{tr}\{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}\} = p$ , and  $\mathbf{M}\mathbf{X} = \mathbf{X}$ . So  $(n - p)s^2 / \sigma^2 = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2 / \sigma^2 \sim \chi^2(n - p)$  by Theorem 4.3.2(a), because  $(\mathbf{I} - \mathbf{M})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$ . With  $\mathbf{B} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ ,  $\mathbf{B}(\sigma^2 \mathbf{I})(\mathbf{I} - \mathbf{M}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{X}^\top - \mathbf{X}^\top \mathbf{M}) = \mathbf{0}$ , so  $\hat{\boldsymbol{\beta}} = \mathbf{B}\mathbf{Y}$  is independent of  $s^2$  (Theorem 4.4.1). Since  $\hat{\boldsymbol{\beta}} \sim N_p(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$ , the variable  $X = (\hat{\beta}_j - b) / (\sigma\sqrt{v_{jj}})$  is  $N((\beta_j - b) / (\sigma\sqrt{v_{jj}}), 1)$ . The statistic is  $X / \sqrt{V / (n - p)}$  with  $V = (n - p)s^2 / \sigma^2$ , and Definition 4.1.3 applies.

**Exercise B4.**

Show that  $F_\alpha(r, s) \rightarrow \chi_\alpha^2(r)/r$  as  $s \rightarrow \infty$ , where  $\chi_\alpha^2(r)$  is the upper  $\alpha$  point of  $\chi^2(r)$ ,

and that the power  $\Pr\{F(r, s, \gamma) > F_\alpha(r, s)\}$  tends to  $\Pr\{\chi^2(r, \gamma) > \chi_\alpha^2(r)\}$ , the power of the test that knows  $\sigma^2$ .

#### 4.4.4.3 C. Going deeper

##### Exercise C1.

Keep the notation of [Theorem 4.4.6](#) and assume the null hypothesis  $\mathbf{P}_0\boldsymbol{\mu} = \boldsymbol{\mu}$ . Put  $a = r - r_0$ ,  $b = n - r$ ,  $U = \mathbf{Y}^\top(\mathbf{P} - \mathbf{P}_0)\mathbf{Y}$ ,  $V = \mathbf{Y}^\top(\mathbf{I} - \mathbf{P})\mathbf{Y}$ , and consider

$$R = \frac{\mathbf{Y}^\top(\mathbf{P} - \mathbf{P}_0)\mathbf{Y}}{\mathbf{Y}^\top(\mathbf{I} - \mathbf{P}_0)\mathbf{Y}} = \frac{U}{U + V},$$

the share of the reduced model's residual sum of squares that the full model removes.

- Show that the numerator and the denominator of  $R$  are *not* independent, and that their correlation is  $\sqrt{a/(a+b)} = \sqrt{(r-r_0)/(n-r_0)}$ .
- Show nevertheless that  $R \sim \text{Beta}(a/2, b/2)$  exactly, that  $R$  is independent of  $\mathbf{Y}^\top(\mathbf{I} - \mathbf{P}_0)\mathbf{Y}$ , and that  $F = (b/a)R/(1-R)$ , so the two tests are the same test.
- Which hypothesis of [Theorem 4.4.2](#) fails for the pair  $\{\mathbf{Y}^\top(\mathbf{P} - \mathbf{P}_0)\mathbf{Y}, \mathbf{Y}^\top(\mathbf{I} - \mathbf{P}_0)\mathbf{Y}\}$ , and why does its failure not damage the exact null distribution?

*Solution.*

- The matrices  $\mathbf{P} - \mathbf{P}_0$  and  $\mathbf{I} - \mathbf{P}$  are symmetric idempotent with zero product (proof of [Theorem 4.4.6](#)), so  $U$  and  $V$  are independent with  $U/\sigma^2 \sim \chi^2(a)$  and  $V/\sigma^2 \sim \chi^2(b)$  by [Theorem 4.4.4](#); the noncentralities vanish because  $\mathbf{P}\boldsymbol{\mu} = \boldsymbol{\mu}$  and  $\mathbf{P}_0\boldsymbol{\mu} = \boldsymbol{\mu}$ . The denominator is  $U + V$ , so  $\text{Cov}(U, U + V) = \text{Var}(U)$  and

$$\text{corr}(U, U + V) = \frac{\text{Var}(U)}{\sqrt{\text{Var}(U)\text{Var}(U + V)}} = \sqrt{\frac{2a\sigma^4}{2(a+b)\sigma^4}} = \sqrt{\frac{a}{a+b}},$$

which is positive: a large  $U$  inflates the denominator too.

- This is Exercise (B4) applied to the independent pair  $(U/\sigma^2, V/\sigma^2)$ : the ratio  $U/(U + V)$  is  $\text{Beta}(a/2, b/2)$  and is independent of  $U + V$ , and  $U/V = R/(1 - R)$  gives  $F = (U/a)/(V/b) = (b/a)R/(1 - R)$ , a strictly increasing function of  $R$ . So rejecting for large  $R$  and rejecting for large  $F$  are the same rule at the same level, and  $\sigma^2$  has cancelled in both.
- Independence would require  $(\mathbf{P} - \mathbf{P}_0)(\mathbf{I} - \mathbf{P}_0) = \mathbf{0}$ , whereas  $(\mathbf{P} - \mathbf{P}_0)(\mathbf{I} - \mathbf{P}_0) = \mathbf{P} - \mathbf{P}_0 \neq \mathbf{0}$ . What the theorems supply is a decomposition of the *denominator* into two independent pieces, one of which is the numerator. Dependence between numerator and denominator is not a problem in itself; what an exact distribution needs is that the pair be a function of independent chi-squared variables, which it is.

##### Exercise C2.

Let  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{V})$  with  $\mathbf{V}$  positive definite and  $\mathbf{X}$  of full column rank  $p < n$ . Write

$$\mathbf{K} = (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1},$$

$$\tilde{\boldsymbol{\beta}} = \mathbf{K} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}, \quad \mathbf{A} = \mathbf{V}^{-1} - \mathbf{V}^{-1} \mathbf{X} \mathbf{K} \mathbf{X}^\top \mathbf{V}^{-1}, \quad Q = \mathbf{Y}^\top \mathbf{A} \mathbf{Y}.$$

- (a) Show that  $Q = (\mathbf{Y} - \mathbf{X} \tilde{\boldsymbol{\beta}})^\top \mathbf{V}^{-1} (\mathbf{Y} - \mathbf{X} \tilde{\boldsymbol{\beta}})$ , that  $\tilde{\boldsymbol{\beta}} \sim N_p(\boldsymbol{\beta}, \sigma^2 \mathbf{K})$  and  $Q/\sigma^2 \sim \chi^2(n-p)$ , and that  $\tilde{\boldsymbol{\beta}}$  and  $Q$  are independent. Deduce an exact  $t$  distribution for  $\lambda^\top \tilde{\boldsymbol{\beta}}$  studentized by  $Q/(n-p)$ .
- (b) Now take the ordinary least squares  $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$  and  $\text{RSS} = \mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$  with  $\mathbf{M} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ . Show that they are independent for every  $\boldsymbol{\beta}$  if and only if  $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ .
- (c) Check that the condition holds for equicorrelated errors,  $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho \mathbf{1}\mathbf{1}^\top$ , in any model whose columns include  $\mathbf{1}$ , and fails for  $\mathbf{X} = \mathbf{1}$ ,  $n = 3$  and  $\mathbf{V} = \text{diag}(1, 1, 4)$ .

*Solution.*

- (a) Since  $\mathbf{Y} - \mathbf{X} \tilde{\boldsymbol{\beta}} = (\mathbf{I} - \mathbf{X} \mathbf{K} \mathbf{X}^\top \mathbf{V}^{-1}) \mathbf{Y}$ , expanding  $(\mathbf{I} - \mathbf{X} \mathbf{K} \mathbf{X}^\top \mathbf{V}^{-1})^\top \mathbf{V}^{-1} (\mathbf{I} - \mathbf{X} \mathbf{K} \mathbf{X}^\top \mathbf{V}^{-1})$  gives  $\mathbf{V}^{-1} - 2\mathbf{V}^{-1} \mathbf{X} \mathbf{K} \mathbf{X}^\top \mathbf{V}^{-1} + \mathbf{V}^{-1} \mathbf{X} \mathbf{K} (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}) \mathbf{K} \mathbf{X}^\top \mathbf{V}^{-1} = \mathbf{A}$ . By Theorem 3.3.1,  $\tilde{\boldsymbol{\beta}}$  is normal with mean  $\mathbf{K} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} \boldsymbol{\beta} = \boldsymbol{\beta}$  and covariance  $\sigma^2 \mathbf{K} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{V} \mathbf{V}^{-1} \mathbf{X} \mathbf{K} = \sigma^2 \mathbf{K}$ . Next, with  $\boldsymbol{\Sigma} = \sigma^2 \mathbf{V}$ , the matrix  $(\mathbf{A}/\sigma^2) \boldsymbol{\Sigma} = \mathbf{A} \mathbf{V}$  is idempotent, because  $\mathbf{A} \mathbf{V} \mathbf{A} = \mathbf{A}$  by the same expansion, and  $\text{tr}(\mathbf{A} \mathbf{V}) = n - \text{tr}(\mathbf{K} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}) = n - p$ ; also  $\mathbf{A} \mathbf{X} \boldsymbol{\beta} = \mathbf{0}$  because  $\mathbf{A} \mathbf{X} = \mathbf{V}^{-1} \mathbf{X} - \mathbf{V}^{-1} \mathbf{X} = \mathbf{0}$ . So  $Q/\sigma^2 \sim \chi^2(n-p)$  by Theorem 4.3.2(b). For independence take  $\mathbf{B} = \mathbf{K} \mathbf{X}^\top \mathbf{V}^{-1}$  in Theorem 4.4.1:  $\mathbf{B} \boldsymbol{\Sigma} \mathbf{A} = \sigma^2 \mathbf{K} \mathbf{X}^\top \mathbf{A} = \mathbf{0}$ , again because  $\mathbf{A} \mathbf{X} = \mathbf{0}$ . Hence  $\lambda^\top (\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta}) / \sqrt{\lambda^\top \mathbf{K} \lambda Q / (n-p)} \sim t(n-p)$ , exactly as in Exercise (B3).
- (b) With  $\mathbf{B} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$  and  $\mathbf{A} = \mathbf{I} - \mathbf{M}$ , Theorem 4.4.1 says that independence for every mean vector  $\mathbf{X} \boldsymbol{\beta}$  holds iff  $\mathbf{B} \boldsymbol{\Sigma} \mathbf{A} = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V} (\mathbf{I} - \mathbf{M}) = \mathbf{0}$ , that is, iff  $\mathbf{X}^\top \mathbf{V} (\mathbf{I} - \mathbf{M}) = \mathbf{0}$ . Transposing, this says  $(\mathbf{I} - \mathbf{M}) \mathbf{V} \mathbf{X} = \mathbf{0}$ , that is,  $\mathbf{V} \mathbf{X} = \mathbf{M} \mathbf{V} \mathbf{X}$ , which is  $\mathcal{C}(\mathbf{V} \mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ . (Necessity is for all  $\boldsymbol{\beta}$ ; for one particular  $\boldsymbol{\beta}$  the covariance may vanish by accident.)
- (c) For the equicorrelated  $\mathbf{V}$  and a model with  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ ,  $\mathbf{V} \mathbf{X} = (1 - \rho) \mathbf{X} + \rho \mathbf{1}(\mathbf{1}^\top \mathbf{X})$ , and each column of  $\mathbf{1} \mathbf{1}^\top \mathbf{X}$  is a multiple of  $\mathbf{1}$ , so  $\mathcal{C}(\mathbf{V} \mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ : ordinary least squares keeps its exact  $t$  and  $F$  theory. For  $\mathbf{X} = \mathbf{1}$  and  $\mathbf{V} = \text{diag}(1, 1, 4)$ ,  $\mathbf{V} \mathbf{1} = (1, 1, 4)^\top$  is not a multiple of  $\mathbf{1}$ , so  $\bar{Y}$  and the residual sum of squares are dependent, and the usual  $t$  statistic has no  $t$  distribution. The condition  $\mathcal{C}(\mathbf{V} \mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$  is the same one under which ordinary and generalized least squares coincide (Chapter 31).